

Syed Abbas Bukhari, PhD

Computational and Systems Biology | Therapeutic Genomics

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Google Scholar: <https://scholar.google.com/citations?user=xxW0dkEAAAAJ>

PROFILE

Computational scientist connecting molecular variation to phenotype through regulatory genomics, single-cell analysis, and pediatric oncology. Leads analyses from study design and data integration through interpretation, reporting, and manuscript preparation. Develops reproducible software and translates genomic evidence into testable target and biomarker hypotheses for collaborative therapeutic research.

CURRENT EXPERIENCE

Staff Scientist, Oncogenomics Section, National Cancer Institute, NIH

2024-present | Bethesda, Maryland | Supervisor: Javed Khan, MD

- Lead computational analysis of a 321,704-cell RMS and muscle-reference atlas; the filtered RMS discovery set includes 104,917 malignant and microenvironmental cells from 21 patients.
- Connect copy-number-informed classification, developmental-state estimates, and consensus expression-program discovery to regulatory interpretation. Current analysis identified 32 metaprograms.
- Project programs into clinically annotated bulk cohorts and evaluate outcomes with model diagnostics and multiple-testing correction. Associations remain exploratory; lead-author manuscript in preparation.
- Contribute computational leadership to a pan-pediatric project integrating consensus regulatory networks, functional dependencies, and compound resources to prioritize candidate vulnerabilities; manuscript in preparation.
- Build documented R/Python workflows for NIH Biowulf and mentor colleagues in analytical design, statistical evaluation, and biological interpretation.

ANALYTICAL CAPABILITIES

Integrative genomics: single-cell and bulk RNA-seq; Seurat, Scanpy, Harmony, CopyKAT, CytoTRACE2; chromatin-informed transcriptional analysis.

Mechanism and target hypotheses: NMF programs; ARACNe-AP, GENIE3, SCENIC/pySCENIC, DoRothEA; integration with functional-dependency and pharmacologic resources.

Statistical evaluation: GSVA/UCell, Cox regression, Kaplan-Meier analysis, proportional-hazards diagnostics, resampling, sensitivity analysis, multiple-testing correction.

Reproducibility and delivery: R, Python, Bash, Bioconductor, SLURM, containers, Git, documented workflows, code review, and collaborator-facing interpretation.

SOFTWARE

SigClust: open-source R workflow for signature-based annotation of existing single-cell clusters with effect sizes and false-discovery-rate control. <https://github.com/bukhariabbas/SigClust>

Research foundation and selected contributions

Postdoctoral Research Fellow, Institute for Genomic Biology, UIUC

February 2020-February 2023 | Urbana, Illinois

Regulatory-network analyses of neurogenomic responses; methodological contributions to the single-cell honeybee brain study in *Nature Ecology & Evolution* (2023).

Bioinformatics Consultant, York University, Canada

2021–2023 | Remote, concurrent consultancy

Bioinformatics analyses for honeybee toxicology, contributing to honeybee health and physiology publications, including *Proteomics* (2026).

Postdoctoral Research Associate, Integrative Biology, UIUC

August 2018-February 2020 | Single-cell and bulk analyses of social-behavior systems and cross-species comparative frameworks.

Research Associate, Institute for Genomic Biology, UIUC

2013–2017 | Earlier research foundation in comparative neurogenomics.

EDUCATION

PhD, Informatics, University of Illinois Urbana-Champaign, 2018. Advisor: Alison M. Bell.

MS, Bioinformatics, University of Illinois Urbana-Champaign, 2012.

BS, Bioinformatics, COMSATS Institute of Information Technology, Pakistan, 2009.

SELECTED PEER-REVIEWED PUBLICATIONS

Traniello IM, Bukhari SA, Dibaeinia P, et al. Single-cell dissection of aggression in honeybee colonies. *Nature Ecology & Evolution*. 2023;7:1232–1244. doi:10.1038/s41559-023-02090-0.

Bukhari SA, et al. Neurogenomic insights into paternal care and its relation to territorial aggression. *Nature Communications*. 2019;10:4437. doi:10.1038/s41467-019-12212-7.

Bukhari SA, et al. Temporal dynamics of neurogenomic plasticity in response to social interactions in male threespined sticklebacks. *PLOS Genetics*. 2017;13:e1006840. doi:10.1371/journal.pgen.1006840.

Akhtar MN, Bukhari SA, et al. POLYAR, a new computer program for prediction of poly(A) sites in human sequences. *BMC Genomics*. 2010;11:646. Equal contribution.

MENTORING AND SCIENTIFIC COMMUNICATION

Mentor colleagues in transcriptomic analysis, regulatory networks, statistical evaluation, and reproducible computing. Prior teaching includes Applied Statistical Methods I and Introduction to Applied Statistics; invited presentations include the University of Missouri and University of Texas at Austin.